1.4 Solving Two-Step Equations

Lesson Introduction

Benchmark	MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical context or real-world context, where all terms are rational numbers.	Learning Targets	• I can solve two-step equations in one variable. • I can relate the solution of an equation representing a real-world context to the context.

	Building On	Working Toward
Benchmark Coherence	MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers. MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.	MA.8.AR.2.1 Solve multi-step linear equations in one variable with rational number coefficients. Include equations with variables on both sides.

Essential Questions	• How do you solve two-step equations? • How do you interpret the solutions within the real-world contexts they represent?
Lesson Narrative	In this lesson, students continue solving two-step equations with rational numbers using the properties of equality. Students extend their knowledge and skills to solve equations that represent real-world contexts.

Component Summary	1.4.1 Warm-Up (~5 min.)	Describe and correct errors in a sample of student work *(SE p. 25)*	1.4.3 Guided Instruction (10-15 min.)	Relate equations and their solutions to real-world contexts *(SE p. 26)*
	1.4.2 Collaboration (10 min.)	Describe and correct errors in samples of student work *(SE p. 25)*	1.4.4 Your Turn! (10 min.)	Practice solving two-step equations *(SE p. 28)*
	1.4.5 Wrap-Up (~5 min.)	Students summarize their learning *(SE p. 29)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: N/A <u>Additional Terms</u>: • Multiplication property of equality • Division property of equality • Subtraction property of equality • Addition property of equality • Symmetric property of equality	• Calculators	• Consider creating and displaying an anchor chart with the properties of equality for students to reference.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may not realize that $-x = -1x$.
Students may struggle to determine what the variable represents in real-world problems.
Students may struggle to determine which property of equality to use in which order.</td><td>Consider providing students with sentence stems or answer choices for constructive response problems.
Consider having no-calculator questions where students turn their calculators over. Teachers should look for and consider opportunities for students to improve their fluency with rational numbers.
Refrain from using the word “cancel.” Instead, say that the inverse operations add/subtract to 0 or multiply/divide to 1.
If time is an issue, then consider assigning Your Turn! for homework.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Compare and Connect
In 1.4.2 Collaboration, students discuss the similarities and differences between Wilma and Maurio’s work. Ask students to connect these comparisons to the properties of equality. These discussions will help students gain confidence and understanding.</td><td>MA.K12.MTR.5.1: Patterns and Structure
This lesson focuses on solving two-step equations. In 1.4.2 Collaboration, students review the structure of student work to complete it and to notice that both methods have the same solution. Students examine the structure of the work for similarities and differences. Students then examine the work of three students. In looking at the structure of the work students fix the errors in the hope of not having similar misconceptions.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">For 1.4.2 Collaboration, if students are struggling to identify the mistake, ask them to solve the problem themselves and then compare their work with the work of the student to help find the mistake.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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1.4.1 Warm-Up: Bell Work

Component Overview: *Students review the properties of equality by describing and correcting errors in a sample of student work. Ask students what the coefficient of the variable is in the original equation. Students may not initially notice that the coefficient is -1.*

1.4.1 Warm-Up: Bell Work

Michaela solved the equation $2.9 - r = -5.11$. Her work is shown.

$$\begin{array}{rcl} 2.9 - r & = & -5.11 \\ -2.9 & & -2.9 \\ \hline r & = & -8.01 \end{array}$$

1. Use substitution to verify if Michaela's solution is correct.
   $2.9 - (-8.01) = -5.11$
   $10.91 \neq -5.11$

2. If Michaela found the correct solution, describe the properties of equality applied. If Michaela did not find the correct solution, describe the error(s) she made.
   Michaela's solution was incorrect. She applied the subtraction property of equality correctly, but she forgot to bring down the negative sign with the variable r. The coefficient of r is -1. To find the correct solution she would have needed to apply the division property of equality and divide both sides of the equation by -1. The correct solution is $r = 8.01$.

Use information from the student responses in the Warm-Up to help determine student understanding of the common error shown in Michaela's work. Consider student understanding to determine if small-group intervention would be helpful.

In the previous lesson, students reviewed the properties of equality. Consider displaying the properties for students to reference throughout the lesson.

1.4.2 Collaboration: Examining Student Work

Component Overview: *Students collaboratively describe and correct errors in samples of student work. Prompt students to analyze each step of the work samples and discuss the mistakes.*

1.4.2 Collaboration: Examining Student Work

1. Wilma and Maurio solved the equation $42 = {}^{-}3x + 15.6$ by applying the subtraction property of equality and the division property of equality. Their work is shown.

Wilma's Work	Maurio's Work
$42 = {}^{-}3x + 15.6$ $42 - 15.6 = {}^{-}3x + 15.6 - 15.6$ $26.4 = {}^{-}3x$ $\frac{26.4}{-3} = \frac{{}^{-}3x}{-3}$ $\underline{-8.8} = x$	$42 = {}^{-}3x + 15.6$ $\frac{42}{-3} = \frac{{}^{-}3x}{-3} + \frac{15.6}{-3}$ ${}^{-}14 = x - 5.2$ ${}^{-}14 + 5.2 = x - 5.2 + 5.2$ $\underline{-8.8} = x$

a. Complete the last step in each student's work. Then, check your work with your partner.

b. Discuss the similarities and differences between each students' work with your partner. Summarize your discussion in the table.

How are the students' work similar?	How are the students' work different?
Answers will vary. *Both students applied the division property of equality.*	*Answers will vary.* *The students applied the properties of equality in different orders.* *Wilma used subtraction first and Maurio used division first.*

The Collaboration introduces to students that it is possible to perform division before subtraction. The method of dividing first may have a greater possibility of error but it can be done. While circulating listen for student confusion on Maurio's work and assure students that his method is viable. During the Guided Instruction make a point of showing students that it is possible to use Maurio's method.

Compare and Connect
Students discuss the similarities and differences between Wilma and Maurio's work. Ask students to connect these comparisons to the properties of equality. These discussions will help students gain confidence and understanding.

1.4.2 Collaboration: Examining Student Work (continued)

Notice that when division property of equality is applied first, each term in the equation must be divided by the same value to maintain equivalence.

2. Several students made errors when solving equations. Their work is shown in the table.

   Work with your partner to find the error in each student's work. Circle the step where the student made the error. Then, work together to complete the table.

Student Work	Describe the Error	Correct Solution
$-6t - 7 = 29$ $+7 \quad +7$ $-6t + 0 = 36$ $\frac{-6t}{6} = \frac{36}{6}$ $t = 6$	*In the second step, the student divided both sides by* 6 *instead of* -6.	$t = -6$
$17 = -3 + 4y$ $+3 \quad +3$ $20 = 4y$ $-4 \quad -4$ $16 = y$ $y = 16$	*In the second step, the student subtracted* 4 *from both sides instead of dividing by* 4.	$y = 5$
$\left(\frac{-2}{1}\right) - 5 = -\frac{1}{2}x^{(-2)} - 3$ $10 = x - 3$ $+3 \quad +3$ $13 = x$	*When applying multiplication property of equality, the student did not multiply all terms on the right side by* -2.	$x = 4$

Think-Pair-Share
Ask students to initially complete question 2 independently and then discuss their work with their partner.

If students are struggling to identify the mistake, ask them to solve the question themselves and then compare their work with the work of the student to help find the mistake.

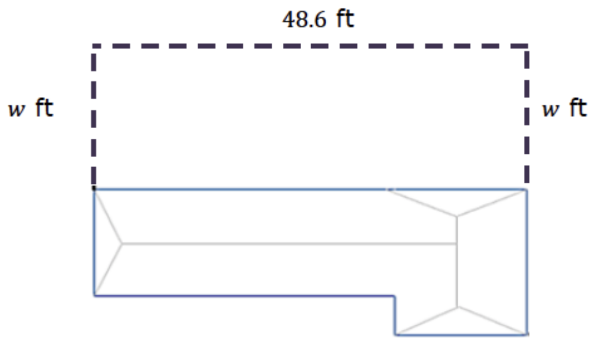
1.4.3 Guided Instruction: Relating Equations and Their Solutions to Real-World Contexts

Component Overview: *Students develop an understanding of how to relate equations and their solutions to real-world contexts. Students solve two-step equations using the properties of equality and describe what the solutions mean in the context of the problems.*

1.4.3 Guided Instruction: Relating Equations and Their Solutions to Real-World Contexts

Two-step equations can be used to represent real-world situations when solving problems.

Dan's family recently adopted a dog, so they are putting in a new fence in their backyard. Dan found fencing on sale at a local hardware store, but the store only had 80 linear feet (ft) of the fencing left in stock. Dan knows the backyard is 48.6 ft long and wants to determine how wide the fenced-in portion of the yard would be if he used 80 linear ft of fencing. The dotted line in the figure shows how the fence will be constructed to enclose the backyard.



1. The equation $48.6 + 2w = 80$ can be used to determine how wide the fenced-in portion of the yard will be.

   a. Solve $48.6 + 2w = 80$. Show your work.

   $48.6 + 2w = 80$
   $48.6 - 48.6 + 2w = 80 - 48.6$
   $2w = 31.4$
   $\frac{2w}{2} = \frac{31.4}{2}$
   $w = 15.7\ ft$

Engage students in a discussion about how the equation was formed. Also, check that students know what each number and what the variable represents.

b. Verify your solution is correct.

$48.6 + 2(15.7) = 80$

$48.6 + 31.4 = 80$

$80 = 80$

c. Explain what the solution means in this context.

The fenced in portion should be $15.7ft$ wide on both sides.

Highlight the importance of checking your answer by substituting your solution back into the equation.

In most countries, temperatures are measured in degrees Celsius, while in the United States, the Fahrenheit scale is used. The equation $C = \frac{F-32}{1.8}$ can be used to convert between these scales, where C is the temperature in degrees Celsius (°C) and F is the temperature in degrees Fahrenheit (°F).

2. The equation $25 = \frac{F-32}{1.8}$ can be used to find the temperature in °F when it is 25°C.

a. Solve the equation to find the value of F.

$$25 = \frac{F - 32}{1.8}$$
$$25 * 1.8 = \frac{F - 32}{1.8} * 1.8$$
$$45 = F - 32$$
$$45 + 32 = F - 32 + 32$$
$$77 = F$$

b. Verify your solution is correct.

$$25 = \frac{77 - 32}{1.8}$$
$$25 = \frac{45}{1.8}$$
$$25 = 25$$

c. Explain what the solution means in this context.

When it is 25°C, it is 77°F; they are the same temperature.

For question 2, notice the whole right side of the equation is being divided by 1.8. Students may try to use the subtraction property of equality first. If so, consider a conversation about why we use the division property of inequality first.

1.4.4 Your Turn!

Component Overview: *Students independently practice solving two-step equations using properties of equality and determining what their solutions mean within the contexts of the problems.*

1.4.4 Your Turn!

1. The basketball team at C.H. Price Middle School in Putnam County, Florida sold candy bars to raise money for new Razorbacks uniforms. The coach spent $\$78.95$ on supplies for the fundraiser and team members sold each candy bar for $\$1.50$. When the fundraiser ended, the team had $\$285.55$ to spend on new uniforms after paying the coach back for the supplies she bought. The equation $1.5c - 78.95 = 285.55$ can be used to represent this situation.

   a. Solve the equation to find the value of c.
   $1.5c - 78.95 = 285.55$
   $1.5c - 78.95 + 78.95 = 285.55 + 78.95$
   $\frac{1.5c}{1.5} = \frac{364.50}{1.5}$
   $c = 243$

   b. Verify your solution is correct.
   $1.5(243) - 78.95 = 285.55$
   $364.50 - 78.95 = 285.55$
   $285.55 = 285.55$

   c. Explain what the solution means in this context.
   The team sold a total of 243 candy bars.

The variable c is not defined in the question 1 description. Students need to determine what the variable means in the context of the problem. Prompt students to underline important information in the problem and ask themselves if the value for each is given or unknown. Also, remind students to check if their answer makes sense in the given context.

2. Solve $-10\frac{1}{3}+\frac{x}{6}=-\frac{8}{3}$ using the properties of equality.
   Then, verify your solution is correct.
   Work:

$$-10\frac{1}{3}+\frac{x}{6}=-\frac{8}{3}$$
$$-10\frac{1}{3}+10\frac{1}{3}+\frac{x}{6}=-\frac{8}{3}+10\frac{1}{3}$$
$$\frac{x}{6}=7\frac{2}{3}$$
$$\frac{x}{6}*6=7\frac{2}{3}*6$$
$$x=46$$

   Check:

$$-10\frac{1}{3}+\frac{46}{6}=-\frac{8}{3}$$
$$-\frac{8}{3}=-\frac{8}{3}$$

In question 2 students need to recognize that the inverse operation of adding a negative is adding a positive. Consider modeling a similar example to support students if they struggle on this problem. Applying the commutative property on the left side of the equation may help students identify the inverse operation.

3. Solve $-9.2=\frac{m+7.22}{4}$ using the properties of equality.
   Then, verify your solution is correct.
   Work

$$-9.2=\frac{m+7.22}{4}$$
$$-9.2*4=\frac{m+7.22}{4}*4$$
$$-36.8=m+7.22$$
$$-44.02=m$$

   Check

$$-9.2=\frac{-44.02+7.22}{4}$$
$$-9.2=-\frac{36.8}{4}$$
$$-9.2=-9.2$$

1.4.5 Wrap-Up: Cool Down

***Component Overview**: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.*

1.4.5 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.

   Answers will vary.